GAGA

In this section, we discuss the GAGA (Géométrie Algébrique et Géométrie Analytique) principle for Berkovich spaces.

Construction 1. Let X be a scheme of finite type over a non-archimedean field k. We can associate to X its Berkovich analytification X^{an}, which is a Berkovich analytic space over k.

The analytification functor $(\cdot)^{\mathrm{an}}$ has the following properties:

- It is compatible with fiber products: for schemes X and Y over k, we have $(X \times_k Y)^{\mathrm{an}} \cong X^{\mathrm{an}} \times_k Y^{\mathrm{an}}$.
- It preserves open and closed immersions.
- It induces a homeomorphism between the underlying topological spaces of X and X^{an}.
- For a coherent sheaf $\mathcal{F}$ on X, there is an associated coherent sheaf $\mathcal{F}^{\mathrm{an}}$ on X^{an}.

Yang: To be revised.

Appendix

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